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RESEARCH ARTICLE

Behavioral Empathy Classification in Dyadic Interactions Using Multimodal Cues

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ABSTRACT Empathy in human–human interactions encompasses nonverbal behaviors that are often overlooked by state-of-the-art empathic perception models, despite extensive progress in the text domain. In this article, we investigate empathy classification in dyadic interactions using audiovisual data, specifically arousal and valence values extracted from facial expressions, head gestures, and textual content derived from speech. We propose a novel cue integration framework that augments precomputed text-based features with video-derived emotional cues and applies them to a pattern-based empathy classifier. Furthermore, we introduce a ranking module that provides interpretability by revealing the relative influence of empathy cues. Our approach is evaluated in two distinct settings: a controlled laboratory environment in which multiple pairs of participants engage in face-to-face conversation, and a real-world school setting involving group interactions among children. Results show that our multimodal empathy classifier generalizes better than deep learning baselines and on par with state-of-the-art large language models across both dyadic and group interaction scenarios, without requiring retraining.

INDEX TERMS Affective computing, empathy, empathy classification, audiovisual data, emotion detection, large language models, LLMs, multimodal classification, pattern classification, text-classification, vision.

I. INTRODUCTION

Empathy is a complex psychosocial phenomenon that involves recognizing and resonating with another person's cognitive and emotional state [1], [2]. Evidence suggests that it has been crucial for the development of human social dynamics, effective communication, and even the emergence of morality and justice [3]. As such, the perception and modeling of empathy enable a deeper understanding of social interaction compared to emotion recognition, making it a highly sought-after target in affective computing research [4], [5]. Significant progress has been made in recent years in empathy prediction and classification. Nevertheless, most approaches have focused on detecting empathy in the text domain [6], while other modalities remain relatively underexplored, despite empathy being conveyed through multiple channels such as facial expressions and body posture [7], [8].

The motivation behind our research is to highlight the importance of nonverbal empathic cues in classification algorithms for dyadic empathy interactions. To this end, we introduce a novel technique that augments text-based empathy classifiers with visual information without requiring additional training. In contrast to previous frameworks [9], our approach allows us to integrate emotional features extracted from facial expressions and head gestures directly.

We focus on the study of behavioral empathy, defined as observable actions that reflect both affective resonance and cognitive understanding of another person's emotional state [1]. This form of empathy is particularly relevant

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because it is externally perceivable and can be annotated through observable behaviors. These behaviors, referred to as *empathy cues*, span multiple modalities such as words of concern (contextual), tone of voice (auditory), and facial expressions (visual) [10], [11]. Such cues characterize behavioral empathy and can be systematically observed and quantified for use in affective computing applications. Previous multimodal empathy classification studies have introduced datasets containing audiovisual information along with baseline methods for comparison [4], [12]. However, none have specifically targeted behavioral empathy. To address this gap, we collected two new datasets of face-to-face dyadic conversations and group interactions in an exchange-based format. These datasets enable us to evaluate our proposed model in multimodal contexts and to explore the impact of contextual and video-based cues on empathy classification.

Our proposed classifier builds upon a pattern-based model proposed by Montiel-Vázquez et al. [11]. Our model is likewise designed for exchange-based interactions and evaluates the empathetic quality of listener responses to topics introduced by speakers. We extend it by using different features [13] and incorporating visual information to enrich the representation of empathy-related cues. Furthermore, we introduce a ranking algorithm for pattern-based classifiers that quantifies the influence of individual features on empathy classification, offering interpretability and deeper insight into the role of each cue. The classifier features extracted from visual information are referred to as *video-based empathy cues*, while those derived from textual data are referred to as *contextual empathy cues*.

Multimodal empathy classifiers are typically developed using either a top-down or bottom-up approach [4], [12], [14]. We introduce a framework for top-down classifiers to integrate the following psychology-based empathy cues: arousal and valence affective vectors (AV), a model of emotions represented by two dimensions [15], and the presence of visual mimicry [1]. This framework allows us to augment precomputed classifiers—such as pattern-based models—as well as large language models (LLMs). In turn, it enhances model performance across multiple interaction scenarios while remaining computationally efficient.

We conduct multimodal classification in two distinct scenarios. The first scenario consists of dyadic face-to-face conversations collected in a controlled laboratory environment. We introduce these interactions as a novel dataset for evaluating behavioral empathy classifiers, named **m**ultimodal **e**mpathetic e**x**changes (**mEX**). To the best of our knowledge, **mEX** is the first behavioral empathy dataset to include synchronized audiovisual recordings. The second scenario involves *The Talking Room*, a multi-group school setting with robot mediators, where children engage in conversations about their preferences and daily habits [16]. Across both settings, we collected 393 exchanges containing multimodal data. While empathy-related datasets are becoming increasingly common, most focus on other forms of empathy—such as self-assessed internal states [4]—or lack audiovisual information altogether [17].

The accuracy of our multimodal model demonstrates that precomputed empathy classifiers can be effectively applied to both dyadic and group interaction scenarios. Moreover, our proposed integration framework enhances the performance of our pattern-based algorithm by incorporating complementary multimodal features. We believe this work encourages further development of multimodal empathy classifiers and contributes to a deeper understanding of the observable cues that shape empathy in human–human interactions.

All prompts, models, datasets, and algorithms required to reproduce the proposed method are available in our public repository.[1]

This article is structured as follows. Section II introduces our definition of empathy. Section III reviews related work. Section IV presents our framework for multimodal empathy classification. Section V describes the concept of *empathy cues* and distinguishes between contextual and audiovisual cues. Section VI details the development of the multimodal empathetic exchanges dataset. Section VII explains our proposed multimodal classifier for empathy detection. Evaluation results are presented in Section VIII. Likewise, we present an implementation of various multimodal baselines, trained and evaluated on **mEX**, in Section IX. Then, we present an in-the-wild group setting assessment of our classifier in Section X. Section XI discusses the limitations of our research, while Section XII examines the implications of our findings and directions for future work. Finally, we conclude in Section XIII.

A. CONTRIBUTIONS

The main contributions of this work are as follows:

1) **Multimodal Feature Augmentation** — We propose a simple integration framework for video-based empathy cues that adapts text-based algorithms for behavioral empathy classification. Our approach enables the augmentation of pre-trained models without requiring additional data collection or retraining to incorporate video-based cues.
2) **Empathy Cue Influence Analysis** — We introduce a ranking algorithm module for PBC4cip [18] to analyze the relevance and contribution of multimodal cues during inference in empathy classification.
3) **Multimodal Empathy Dataset** — We present *mEX*, a multimodal empathetic exchanges dataset designed for evaluating multimodal empathy classifiers.

II. EMPATHY DEFINITION

The first step of our research is to clarify the definition of empathy in which we base our work. This is necessary as the characteristics of empathy vary depending on their theoretical basis [1], [2].

[1] https://github.com/christianharu/MultimodalEX

The definition of empathy is still heavily discussed in psychological literature [1]. This has led to differing definitions for the same concept in affective computing applications [19], [20], [21], [22], [23]. However, empathy is often conceptualized as a psychological construct that can be either a state or a trait. This construct is multi-dimensional [2] and its components are hierarchical [24]. The latter characteristic refers to the fact that the lower forms of empathy are combined with more complex components to obtain higher levels of empathy [25].

We adopt the definition of empathy presented by Clark et al. [1]. This definition is grounded in a multidisciplinary body of research and has been previously used in affective computing [11]. It states that empathy comprises three dimensions: affective, cognitive, and behavioral empathy. Affective empathy refers to experiencing an emotional state similar to the one expressed by another person, which is considered the lowest form of empathy [25], [26]. Cognitive empathy entails understanding another person's affective state and situation, requiring more complex mental processes that build upon affective empathy [7]. Finally, behavioral empathy refers to the state or trait of engaging in verbal and nonverbal behaviors that demonstrate affective and/or cognitive empathy. As such, this latter dimension encompasses the previous ones and occupies a higher position in the hierarchical order.

The hierarchical and multi-dimensional structure of empathy allows us to build a top-down approach for classifying behavioral empathy in dyadic conversations. Further details for our approach along with important concepts can be found in Section V.

III. RELATED WORK

This section discusses key areas relevant to our research. We provide an overview of text-based and multimodal empathy classification approaches, describe the datasets commonly used in previous studies, and summarize applications of empathy classifiers in different contexts.

A. EMPATHY CLASSIFIERS

A large number of works have addressed empathy classification across diverse settings. These range from algorithms performing binary classification [5], [20], [27], [28] to multi-label models [11], [29], [30]. The most recent comprehensive survey on empathy detection, encompassing both classification and regression tasks, was presented by Hasan et al. [6]. For a complete overview of the field, we refer readers to their work.

Empathy classification shows consistent trends despite the varied conceptualizations of empathy. The authors in [6] note that most approaches to empathy classification rely on text sequences. Consequently, the state-of-the-art (SOTA) has trended toward transformer-based deep learning algorithms for this task. This direction has been explored across various textual sources, including essays [20], [31], asynchronous social media posts [28], [30], [32], [33], synchronous text conversations [11], [34], and other text modalities [35]. The general strategy involves fine-tuning pre-trained models on labeled data or on instances annotated through heuristic methods [5]. As a result, most SOTA algorithms are not domain-specialized and differ in their implementation across datasets. Furthermore, variations in label definitions and training corpora make comparisons between them difficult. Beyond these approaches, recent developments have shown that LLMs can outperform traditional deep learning models even without specialized training or fine-tuning [12], [36], [37]. In this work, we consider both models pre-trained on empathy datasets, and LLMs as baselines for text-based empathy classification.

Explainable algorithms are seldom explored in empathy classification. They provide interpretable information about the underlying reasoning behind classification decisions. As a result, they offer valuable insight into empathetic behavior [29]. Notable examples include the EPITOME framework [30] and its extensions [11], [38]. In our work, we adopt the PBC4cip algorithm [11], [18], [29], a contrast pattern-based classifier that extracts influential patterns for each label to classify instances while accounting for class balance weights.

Following the approach described in [11], we classify empathy at the exchange level to better capture its nature as a phenomenon that emerges between two or more individuals [28]. Our modified algorithm achieves state-of-the-art performance on the EmpatheticExchanges (EX) dataset and further enables the identification of influential empathy cues within each exchange, improving interpretability and supporting behavioral analysis.

B. MULTIMODAL EMPATHY CLASSIFIERS

Several studies have combined multiple modalities to enhance empathy classification [4], [12], [14], generally following either top-down or bottom-up approaches.

Bottom-up methods are the most common strategy for incorporating audiovisual data as input for classifiers. This line of research focuses on developing novel fusion techniques and algorithms [39]. For instance, the works by Tan et al. [40] and Tavabi et al. [41] demonstrate the integration of multimodal facial features [42] and audio features [43] for empathy prediction. Most approaches employ Long Short-Term Memory (LSTM) networks [44] to fuse multiple modalities [6]. For example, Chen and Li [39] propose a classification algorithm based on a novel LSTM architecture that fuses video, audio, and text modalities. Bottom-up methods have the advantage of being relatively easy to implement; however, they do not provide interpretable information about the classification process.

Top-down methods contrast with the previous approach by focusing on the most common empathy cues identified in psychological research [11]. Since empathy is a complex construct with multiple components, these methods are best

suited for extracting interpretable cues from audio and video signals. For example, Pan et al. [45] employ interpretable feature extraction for binary empathy classification in the context of online instruction. Similarly, the work in [46] combines arousal and valence values extracted from audio with semantic features from text.

Our multimodal empathy framework follows a top-down approach, as all contextual and visual-based cues are explicitly designed. Compared to other methods, we employ a comprehensive set of visual cues drawn from the literature and a simple integration framework to combine modalities. To the best of our knowledge, this framework has not been previously applied to leverage advances in empathy classification for multimodal domains. It provides the advantage of augmenting trained text-only models without requiring model retraining.

C. EMPATHY DATASETS

Several text-only empathy datasets have been developed across diverse contexts [21], [47], [48]. Among these, EmpatheticDialogues (ED) [17] and its extensions [29] have become widely used benchmarks in affective computing. In this work, we train our text-only empathy classifier on an extended version of EX [11], a successor to ED that includes manually annotated labels for behavioral empathy.

In the multimodal domain, OMG-Empathy [4] and EmpathicStories++ [12] provide audiovisual data capturing empathetic reactions to personal narratives. EmpathicStories++ is currently the largest multimodal empathy dataset, featuring over 53 hours of recordings collected during an in-the-wild deployment involving a social robot. In this setting, participants listened to personal stories shared through the robot and were subsequently asked to describe their emotional reactions and self-evaluate their level of empathy toward the storyteller. Both OMG-Empathy and EmpathicStories++ rely on self-reported empathy scores, which primarily capture internal emotional states. Consequently, a gap remains in the state-of-the-art regarding observable empathy, limiting the development of large-scale models for behavioral empathy classification using audiovisual data.

In contrast to previous datasets, we introduce mEX, a dataset centered on behavioral empathy, emphasizing externally observable empathic actions rather than self-assessments. Annotations were conducted by expert raters based on perceivable verbal and nonverbal cues, providing a more objective ground truth for training and evaluating empathy classifiers. This design supports the development of models that are better suited for real-time applications, where internal self-reflection data are unavailable. We made this dataset available in our repository to contribute for future research.

D. APPLICATIONS

One promising application of empathy classifiers—especially when combined with empathetic text generation algorithms—is the development of agents capable of adapting their level of expressed empathy [13]. For example, Arzate et al. [34] propose a social robot that modulates its empathetic responses based on the perceived empathy level of its human interlocutor. Empathy classifiers can also serve as proxies for assessing interaction quality in human-robot interaction, as higher levels of empathic exchange are generally associated with more effective communication [49], [50].

In addition, empathy classifiers can support the generation of more human-like empathetic text by serving as annotation tools or evaluation components. For instance, the EPITOME framework [30] can be used to automatically annotate datasets with empathy labels, which can then be employed to fine-tune LLMs for empathetic response generation. Such agents have been proposed for healthcare forums [51], where empathic agents have demonstrated higher empathic performance than physicians [52], and for educational contexts, where empathic robots show potential for fostering meaningful discussions. Likewise, empathic agents have been designed to teach individuals how to respond with empathy across diverse scenarios [53], [54]. Access to accurate automatic evaluation of empathic performance is critical, as the use of inappropriate or inaccurate empathic responses can reduce user trust in these systems [55]. As such agents are increasingly deployed across domains, it becomes imperative to develop reliable methods for evaluating and labeling empathic performance.

Finally, as proposed by Hasan et al. [6], empathy classifiers can also be employed to develop automatic moderation strategies for social media. These tools are particularly valuable in online platforms designed to provide emotional support [30], where detecting and promoting empathetic communication is essential [28].

IV. EMPATHY CLASSIFICATION FRAMEWORK

This section provides an overview of our framework for classifying behavioral empathy in multimodal conversations. We first describe the empathic scenarios used to investigate empathy classification, then outline the data processing steps required to integrate the recordings into our datasets. Finally, we define the performance metrics employed throughout the study. Figure 1 illustrates the overall architecture of the framework, which includes the sensing and perception modules, interaction setup, extracted visual cues, datasets, labeling procedure, and the multimodal empathy classifier.

A. EMPATHETIC SCENARIOS

To investigate behavioral empathy, we analyze multimodal conversations composed of paired utterances, referred to as *exchanges* [11]. Each exchange consists of a *speaker* who initiates the dialogue and a *listener* who responds. Our objective is to classify the degree of empathy exhibited by the listener in their response [28].

Behavioral empathy encompasses both verbal and nonverbal communication, yet most prior research has focused

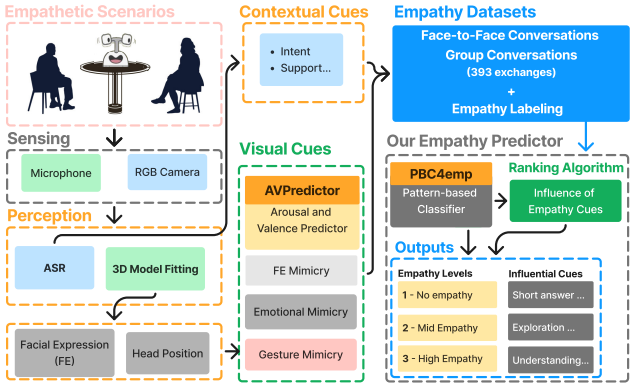


FIGURE 1. Overview of our multimodal empathy framework used to construct the multimodal empathetic exchanges (mEX) dataset.

exclusively on linguistic signals [11]. To address this limitation, we designed two complementary scenarios for empathy prediction.

The first scenario involves dyadic, face-to-face interactions in which participants discuss emotionally charged situations. This setting mirrors the characteristics of the Empathetic-Dialogues dataset [17] but introduces a physical, in-person component that enables the analysis of nonverbal cues such as facial expressions and gestures.

The second scenario employs the social robot Haru [56] in an intercultural group context [16]. In this case, students from different countries engage in video-call discussions about everyday topics such as hobbies, school life, and pop culture. We focus on exchanges between members of different groups, treating each pairwise interaction as a distinct empathy exchange.

We collected a new dataset for the first scenario and utilized an existing one for the second. Together, these datasets enable the validation of our empathy classification framework across both controlled laboratory and real-world cross-cultural settings.

B. EMPATHY DATA AND PROCESSING

Each interaction is recorded using a single microphone and two RGB cameras capturing the participants' upper bodies. Audio is transcribed automatically, and facial expressions are modeled using EMOCA [57] to obtain 3D morphable representations. The resulting multimodal data—comprising transcripts, videos, and extracted empathy cues—are compiled into an empathy dataset designed for evaluation rather than training.

Each exchange is manually annotated using the 3-Level Empathy Score (3-ES) [11], which is simplified into a binary scheme distinguishing between low and high empathy responses. The annotated data are used to evaluate our multimodal empathy classifier, which integrates both contextual and visual cues. Specifically, facial coefficients extracted via EMOCA are combined with text-based features from pre-trained language models to predict empathy levels and estimate cue importance. Further implementation details are provided in Sections V, VI, and VII.

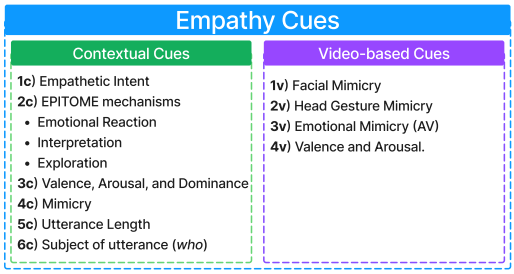


FIGURE 2. Contextual and video-based cues obtained from interactions for empathy classification.

C. PERFORMANCE METRICS

We evaluate our models using **acc**uracy (acc), the **c**loseness **e**valuation **m**easure (CEM) [58], and macro-averaged **F1**-score (F1), **p**recision (p), and **r**ecall (r). The corresponding results are reported in Sections VIII, IX, and X.

V. PERCEIVING EMPATHY CUES

Empathetic behavior includes mirroring mechanisms such as facial expressions, mannerisms, and gestures. It also encompasses empathetic communication, including verbal expressions of understanding, question-asking, and nonverbal agreement (e.g., head nodding). Although these behaviors are not necessarily *indicators* of behavioral empathy, they can serve as useful *predictors* of this phenomenon [1]. Montiel-Vazquez et al. [10], [11] refer to these actions and concepts adjacent to empathetic behavior as *empathy cues*. Extracting empathy cues from dyadic exchanges enables the use of top-down features in a classifier to obtain an empathy label.

Empathy cues are applied in a wide range of affective computing applications. In conversational artificial intelligence, empathy-related concepts such as intent [59] or communication mechanisms [30] are often integrated into the generation pipeline to enhance empathetic capabilities [10]. Likewise, cues are incorporated into empathy classification models through elements such as emotion recognition, emotional signals, and sentiment analysis [6]. Previous work by Lee et al. [38] shows that most empathy classifiers primarily rely on surface-level features. Therefore, we aim to gain deeper insight by analyzing the influence of different empathy cues on classification outcomes.

We base our work on previous studies that incorporate empathy cues into classification [11]. However, we extend this approach by exploring different methods for extracting textual cues and by incorporating cues beyond verbal communication. We present the full scope of empathy cues used in this article in Figure 2.

A. CONTEXTUAL CUES

Text-based empathy cues have been applied to both empathy detection [6] and text generation [60], where they have been shown to improve performance in both domains. We use the contextual cues identified in [11] and [13] to classify the level

of empathy per exchange in our dataset, with multimodal modifications. We select these cues because their contribution to state-of-the-art performance on the EmpatheticExchanges dataset. Also, their use in predicting when someone is seeking or providing empathy in dyadic interactions [13].

Our set of contextual cues includes: the intent of an interlocutor when providing an empathetic response (**1c**) [59], [61]; communication mechanisms used to convey mental health support (**2c**) [30]; the sentiment (**3c**) and emotion of utterances (**4c**) [29]; the presence of emotional mimicry (**5c**) [25], [62]; the number of words in each utterance of the exchange; and (**6c**) the main subject of the utterance [28]. We obtain contextual features using methods from Montiel-Vazquez et al. [11] and Arzate et al. [13].

For **1c, 2c, 3c** and **6c** we use the implementation found in [13] using the *LLaMA3.3* open-source LLM as a classifier. The reason for using this model for these cues is that LLMs, particularly those open-source, have been shown to generalize well to unseen tasks related to affective computing [63], [64] and sentiment analysis with complex contexts [65]. Likewise, they have been observed to work well in Human-Robot Interaction. To ensure that the results are reliable, especially those for **1c** and **2c**, we provide the model with an optimized prompt that contains the description of the construct and examples. We enhance reliability by instructing the model to provide reasoning for the classification label obtained. We obtain a label across 8 intent categories plus a neutral class for **1c** [59], [61]. **2c** extracts the Emotional Reaction, Interpretation, and Exploration mechanisms from the EPITOME framework across 3 communication levels [30]. **3c** outputs a positive, neutral, or negative sentiment label for speaker and listener utterances. Meanwhile, **6c** outputs a label that describes whether the main subject of the utterance is the person speaking (*I*), the conversational partner (*You*), or another person (*other*) [28].

For **4c** and **5c**, we use a dimensional emotion model [66] and obtain arousal, valence, and dominance scores for each utterance using the NRC_VAD lexicon [15]. This dimensional model allows us to compute a binary emotional mimicry label using cosine similarity, which is set to 1 when the similarity exceeds 0.5. Previous work used 0.7 as to select vectors in the same quadrant of the affective space given by Arousal and Valence [11]. However, we use more lenient threshold based on equally dividing in four parts the possible range of cosine similarity $[-1,1]$ and selecting the closest quarter as mimicry. The emotions presented by both participants should be similar but not completely equal on account of differences in expression such as behavioral expressivity or comfort expressing one's emotion or reaction [67].

The final selection of cues covers a wide range of characteristics that make up empathetic communication. We apply this method to the entire EmpatheticExchanges dataset and release it for future use.

B. VIDEO-BASED CUES

Visual empathy cues play a crucial role in perceived behavioral empathy [60], particularly through the outward expression of emotions and mimicry mechanisms [1]. These phenomena are among the most fundamental predictors of empathy [26]. Moreover, there is evidence suggesting that they represent the only empathy cues that can be reliably obtained from visual data, as opposed to neural signals or linguistic communication [68].

To leverage these cues, we develop a method that enhances empathy detection models by incorporating mimicry mechanisms derived from facial expressions (**1v**), head gestures (**2v**), and emotional reactions (**3v**) per exchange. Additionally, we extract emotional information from participants' facial expressions using valence and arousal vectors for each utterance (**4v**).

Although other video-based cues exist (e.g., hand gesture mimicry), we focus exclusively on upper-body signals. This choice reflects both the seated nature of our experimental setup and our objective to employ only non-intrusive sensing technologies.

In the remainder of this section, we describe the algorithms used to compute our video-based cues, using the facial expression and head position coefficients from EMOCA [57] as input.

1) EMOTIONAL DISPLAY

We follow the implementation proposed by Arzate et al. [69], which uses sequences of facial expression coefficients extracted by EMOCA [57] to estimate arousal and valence (AV) values. The emotional representation produced by this method, denoted as (**4v**), consists of arousal and valence scores, each ranging within $[-1, 1]$.

In the EMOCA-based method described in [69], the required input is a cropped face image. We use BlazeFace [70] for face detection and cropping, as it produces high-quality results compatible with EMOCA's input format. The cropped face is then processed by EMOCA [57], which outputs several coefficient vectors corresponding to facial expression, head pose, shape, and texture. The method uses only the facial expression coefficients, represented as a time-dependent vector $\mathbf{F}(t) \in \mathbb{R}^{50}$, where each component typically lies within the range $[-4, 4]$.

To compute AV values, the model requires a temporal sequence of facial expressions. Specifically, we input a sequence of $S = 100$ frames (corresponding to two seconds of video). The full sequence, denoted as $\{\mathbf{F}(t)\}_{t=1}^{S}$, is passed to a bi-directional gated recurrent unit model $\mathcal{M}$ that maps the sequence to the AV space:

$$(a, v) = \mathcal{M}\left(\{\mathbf{F}(t)\}_{t=1}^{S}\right), \quad \text{where } (a, v) \in [-1, 1]^2 \tag{1}$$

The final output is a pair of continuous values representing the predicted arousal and valence levels for the given expression sequence.

We compute arousal and valence (AV) values for each video frame of both the speaker and the listener during each utterance. Let the AV time series for an utterance be represented as:

$$\{(a_t, v_t)\}_{t=1}^{T} \tag{2}$$

where $(a_t, v_t) \in [-1, 1]^2$ denotes the arousal and valence at time frame t, and T is the total number of frames in the utterance.

To generate a stable AV value that characterizes the utterance, we fit a step function to the AV signal over time. Specifically, for each dimension (arousal and valence), we model the signal using a piecewise constant function:

$$s(t; \theta) = \begin{cases} c_1 & \text{if } t \in [1, \tau) \\ c_2 & \text{if } t \in [\tau, T] \end{cases} \tag{3}$$

where θ represents the parameters of the step function. We fit this function using the `curve_fit` method from the SciPy library, applied separately to the arousal and valence sequences.

After fitting, we compute the mean of the resulting step function to obtain a single AV estimate for the utterance:

$$\bar{a} = \frac{1}{T}\sum_{t=1}^{T} s_a(t; \theta_a), \quad \bar{v} = \frac{1}{T}\sum_{t=1}^{T} s_v(t; \theta_v) \tag{4}$$

where $s_a(t; \theta_a)$ and $s_v(t; \theta_v)$ are the fitted step functions for arousal and valence, respectively. The final AV representation of the utterance is the pair $(\bar{a}, \bar{v}) \in [-1, 1]^2$.

2) MIMICRY

We limit the analysis of mimicking behavior to a fixed temporal window of 3 seconds, based on theoretical and empirical findings [71], [72]. This window corresponds to $N = 150$ frames (assuming a 50 fps frame rate) and is used to detect short-term behavioral alignment between the speaker and the listener. In other words, if both participants demonstrate the same behavior within 3 seconds of each other, we consider that the exchange presented mimicry.

To compute mimicry of type (**1v**) and (**2v**), we follow a three-step procedure: (1) we align the expressive signals from the speaker and listener within the 3-second window to maximize their likeness, (2) we compute their similarity, and (3) binarize the result based on an empirical threshold.

Let $\{\mathbf{x}_t\}_{t=1}^{N}$ and $\{\mathbf{y}_t\}_{t=1}^{N}$ denote the time series of EMOCA-based signals from the speaker and listener, respectively. For (**1v**) mimicry, these signals correspond to the *facial expression coefficients*, while for (**2v**) mimicry, they correspond to the *head pose coefficients*. The head pose coefficients are represented by a real-valued vector in $\mathbb{R}^{3k+3}$, where $k = 4$ corresponds to four joints (neck, jaw, and both eyeballs) and a global head rotation.

To capture dynamic behavior, we compute the first-order temporal difference of each signal:

$$\Delta\mathbf{x}_n = \mathbf{x}_n - \mathbf{x}_{n-1}, \quad \Delta\mathbf{y}_n = \mathbf{y}_n - \mathbf{y}_{n-1}, \quad n = 2, \ldots, N \tag{5}$$

In step (1), we align the speaker and listener signals using the cross-correlation function from the NumPy package. This function identifies the optimal lag τ^* that maximizes their similarity:

$$\tau^* = \arg \max_{\tau \in [-\tau_{\max}, \tau_{\max}]} \sum_{n=1}^{N-|\tau|} \Delta\mathbf{x}_n \cdot \Delta\mathbf{y}_{n+\tau} \tag{6}$$

We align both signals under the assumption that the listener will mimic similar behavior to the speaker with a delay in time. Therefore, we would observe an optimal shift within the 3-second window in which both would display roughly the same characteristics. The listener's signal is time-shifted by τ^* to align with the speaker's.

For step (2), we compute the cosine similarity between the aligned derivative signals. This similarity is calculated separately for each mimicry type: (**1v**) using facial expression derivatives and (**2v**) using head pose derivatives. Facial expressions vary, and nodding cannot be perfectly synchronized between two people; therefore, we track the derivatives of facial expression and head pose for each participant.

In step (3), we apply a threshold to binarize the mimicry outcome. If the similarity exceeds 0.01, we classify the behavior as mimicking:

$$\text{mimicry}_{\mathbf{nv}} = \begin{cases} 1 & \text{if } \text{sim}_{\mathbf{nv}} > 0.01 \\ 0 & \text{otherwise} \end{cases} \quad \text{for } \mathbf{nv} \in \{\mathbf{1v}, \mathbf{2v}\} \tag{7}$$

We empirically selected the 0.01 threshold using manually labeled exchange-level binary mimicry data. First, we obtained a subset of mEX consisting of a pilot study involving 10 dyadic conversations, each comprising two exchanges. Then, we annotate the following behaviors: nodding, smiling, and frowning. These are selected based on the definitions of empathy [1] and empathic behaviors [7]. If any of these behaviors are displayed by both participants during an exchange, we consider that exchange to display mimicry. Subsequently, we adjust the similarity threshold from 0 to 1 in increments of 0.01 and record the accuracy on mimicry labels produced by the algorithm per exchange. As shown in Figure 3, this threshold produced the best results.

This procedure is applied independently to each 3-second window across the interaction to assess the presence of facial (**1v**) and head-based (**2v**) mimicry. If at least one window within an exchange exhibits a positive mimicry classification, we consider that the entire exchange contains that type of mimicry. The total duration of the exchange is T frames. Formally, for each mimicry type $\mathbf{nv} \in \{\mathbf{1v}, \mathbf{2v}\}$, we define the exchange-level mimicry label as:

$$\text{mimicry_exchange}_{\mathbf{nv}} = \begin{cases} 1 & \text{if } \sum_{w=1}^{W} \text{mimicry}_{\mathbf{nv}}^{(w)} > 0 \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

where $\text{mimicry}_{\mathbf{nv}}^{(w)}$ denotes the mimicry detection result for window w, and W is the total number of non-overlapping

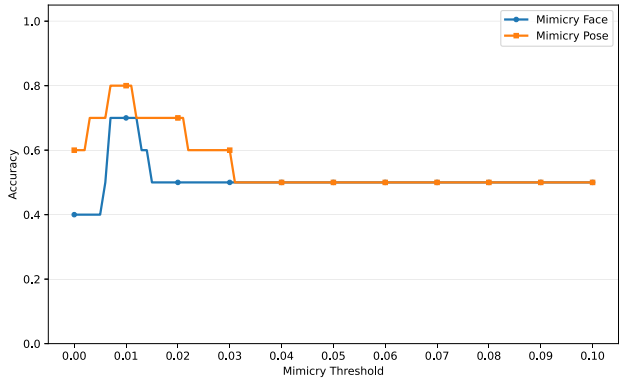


FIGURE 3. Iterative process for selecting the mimicry-threshold hyperparameter based on accuracy.

3-second windows in the exchange (i.e., $W = \lfloor T/150 \rfloor$ for a frame rate of 50 fps).

Finally, to find (**3v**), we compute the cosine similarity of the AV vectors from (**4v**) (the result from Equation 4). If the result is > 0.5, we consider that there is emotional similarity; otherwise, we do not. The threshold selection follows the criteria used in Subsection V-A.

VI. MULTIMODAL EMPATHETIC EXCHANGES

The **m**ultimodal **e**mpathetic e**x**changes (mEX) dataset comprises synchronized video and audio recordings of semi-structured conversations between a *speaker* and a *listener*. The user study for data collection is reviewed and approved by our institution's ethics committee, and all participants provide written informed consent. Figure 4 illustrates the laboratory setup used during data acquisition.

To guide the flow of the conversations, we design prompts for the *speakers* that present real-life situations and their corresponding emotional contexts. Speakers use these prompts as a guide to improvise the conversation, and we encourage them to relate their own experiences to elicit more natural and emotionally grounded responses. Each emotional context corresponds to one of the eight basic Plutchik emotions [73]: joy, trust, fear, surprise, sadness, disgust, anger, and anticipation. The prompts are designed to evoke these contexts and encourage emotional engagement from both participants.

The role of the *listener* is to respond to the speaker at different empathy levels. To obtain natural responses across empathy levels, we explain to the listeners the intended relationship they have with the speaker for each conversation [74]. In other words, listeners role-play based on who they are talking to. The relationships vary between strangers, coworkers, and close friends or family members, corresponding to expected empathy levels of 1, 2, and 3, respectively.

To annotate turn-taking behavior, participants use foot pedals as binary switches. They are instructed to press the pedal once before they begin speaking and press it again upon finishing. This procedure allows us to precisely label speaker turns throughout the interaction.

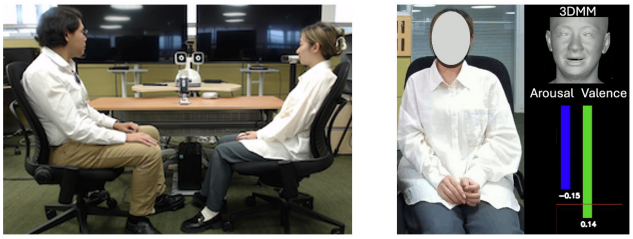


FIGURE 4. Overview of the mEX experimental setup (left) and an example of a participant's 3DMM-based facial representation with the corresponding arousal and valence values (right).

A. TASK

The task for the *speakers* was to initiate an empathetic conversation. The *listeners* followed the conversation flow based on the speakers' input for at least 2 exchanges. Each of the 3 speakers was required to complete sets of 8 conversations with 6 different listeners, resulting in a total of 144 conversations. However, the data of one listener was lost, which necessitated the recruitment of a speaker to carry out a single set of 8 conversations without affecting the total.

B. PARTICIPANTS

The *speakers* were four in total (3 male). Three were authors of this article. Two speakers were from Canada and two from Mexico with different ethnic characteristics. Age distribution consisted of three aged 19–30 and the other 30–40. All of them were able to speak english as native or fluent levels. The Canadian male speaker was recruited to compensate for corrupted or lost data.

The 18 *listeners* (4 female) volunteered for the experiment and were recruited through institutional mailing lists. All participants had prior experience with robots and came from multicultural backgrounds. Five were native English speakers, while the remaining were fully proficient and used English daily. Nine listeners were aged 30–40, and the remaining nine were 19–30.

One listener (male, 30–40) repeatedly broke character, affecting conversation quality. Consequently, all of his conversations were excluded from the dataset.

C. SENSING AND PERCEPTION

We record the conversations using one microphone and two RGB cameras that capture the upper body of each participant. The audio is transcribed automatically using speech recognition, and EMOCA is employed to generate a 3DMM representation of each participant's face [57].

The recorded data are then processed using the algorithms for our empathy cues. We refer to this stage as *perception*, as it allows us to extract both contextual and visual features that are useful for empathy classification (see Section V).

D. DATA COLLECTION

We conducted the data collection in a controlled laboratory setting. Figure 4 (left) shows our robot mediator, Haru, delivering instructions to the participants.

Figure 4 (right) illustrates the types of data we collected, including a 3DMM representation of each participant's face and the corresponding predicted arousal and valence values. The microphone recorded the audio of the conversations, while the robot used a speaker to deliver verbal instructions and prompts. We used the robot to introduce the experiment instructions to avoid potential bias related to the ethnicity or characteristics of a human instructor.

We began by explaining the listeners' role and our definition of empathy, encouraging them to adjust their empathy levels based on their relationship with the speaker. They watched a sample four-exchange conversation and were encouraged to share personal experiences for a natural flow. After addressing any questions, we started the experiment. The introduction lasted approximately $\approx$ 10 minutes.

At the start of each conversation block, Haru welcomed the participants, reminded them of the experiment's goal, and provided an emotional context (e.g., anger) for the speaker and a relationship type (e.g., close friends) for the listener. Haru addressed them by name, displayed a related animation (e.g., smiling for ''joy''), and then observed the speaking participant without engaging verbally. This design kept the interaction dyadic between the humans. At the end, Haru thanked the participants. On average, recording 8 conversations per participant took approximately $\approx$ 35 minutes.

For each conversation, we captured two 1080p resolution videos (one per participant) at 50 fps, along with a two-channel audio file with a sampling rate of 48 kHz.

E. DATA POST-PROCESSING

Each exchange in the mEX dataset included a dialogue transcript, text-based empathy cues, video-based empathy cues, the emotional context of the conversation, and frame-level features such as arousal, valence, facial expression, and head position coefficients. These multimodal features enabled the reconstruction of conversational scenes while preserving participant anonymity through the use of 3DMM representations.

To ensure comparability with the EX dataset, we applied a length-based filter to maintain consistent utterance sizes. We set the mEX dataset utterance maximum length to the ones found in EX. Specifically, we removed conversations in which the speaker or listener utterances exceeded 127 and 103 words, respectively. Using this heuristic, 17 conversations were excluded. The final mEX dataset comprised 127 conversations, totaling 310 exchanges.

F. EMPATHY ANNOTATION

We use an ordinal metric for behavioral empathy, introduced in [11]: the 3-level empathy score (3-ES). This metric defines three hierarchical levels based on the number of elements associated with behavioral empathy. The lowest level is 1, and the highest is 3.

The 3-ES is based on psychological [24], [26] and affective computing research [11], [75]. The lowest level (**1**) represents zero empathy and basic affective empathy elements, such as motor mimicry. The second level (**2**) includes interactions containing affective and cognitive elements in isolation. Finally, the third level (**3**) represents true perspective-taking, integrating cognitive and affective features and an intent to improve or share the emotional state.

Although the 3-ES metric offers a robust framework for empathy classification, we focus on binary classification, which has shown relevance in prior work [6]. To this end, we propose a binary metric with two levels: (**1**) little to no empathetic behavior, and (**2**) empathetic behavior. The latter category encompasses responses that involve at least affective self-regulation and extend beyond emotional contagion, such as expressions of empathetic concern [7]. Notably, this second level corresponds to levels **2** and **3** of the original 3-ES scale. The annotation procedure is described in detail in Section VI.

We obtained empathy labels from two annotators with experience. First, they took the Empathy Quotient (EQ) questionnaire [76] to ensure they did not present deficits in their empathetic abilities. This questionnaire was developed to identify a lack of empathy as a feature of psychological conditions, and it is useful for measuring when a person is incapable of empathy using a single score [29]. Our annotators presented scores of 52 and 63 out of a maximum of 80. These were above the recommended threshold of 30. Therefore, there was no reason to doubt their capabilities in identifying empathy. Subsequently, we trained the annotators by providing relevant definitions and examples with accurate empathy levels [29]. They were instructed to label the behavioral empathy of the listener at the exchange level.

The annotation procedure is iterative. Independent labeling is carried out. Then, there is a discussion to explore the reasoning for discrepancies in the ratings. Subsequently, rounds of labeling are repeated until a consensus is reached. In our work, it was only necessary to perform annotation twice. The first round of 3-level annotations presented a Cohen's Kappa statistic [77] of 0.23. With 2-level annotation, the Cohen's Kappa statistic was 0.38. These were not indicators of strong agreement. We believe that this highlights the difficulty of annotating behavioral empathy. While it is a concept that can be observed, there is inherent subjectivity that requires additional consideration. We also acknowledge that this might lead to label noise. For our work, the next annotation round reached complete consensus after discussion, which allowed us to obtain a behavioral empathy label for each exchange.

We present metrics of annotators against the final result consensus label in Table 1.

The empathy score distribution of 3-ES in mEX is 150 for level 1, 79 for level 2, and 81 for level 3.

TABLE 1. Annotators original labeling against final consensus. CEM = closeness evaluation measure, and F1 = F1-score.

2-level					
Annotator	Accuracy	CEM	F1	Precision	Recall
A	0.845	-	0.844	0.851	0.847
B	0.838	-	0.836	0.846	0.836
3-level					
Annotator	Accuracy	CEM	F1	Precision	Recall
A	0.732	0.813	0.675	0.686	0.677
B	0.725	0.819	0.704	0.702	0.715

G. DATASET STRUCTURE

Our dataset comprises conversational exchanges between a speaker and a listener, where each row corresponds to one exchange. The dataset contains 310 rows and 29 columns. We provide an identifier in column `id` that corresponds to the participant and the conversation for each exchange. Each entry includes the utterances from both interlocutors (`speaker_utterance`, `listener_utterance`) and a set of linguistic and affective features. Columns prefixed with `s_` and `l_` indicate features associated with the speaker and listener, respectively, such as dimensional emotion scores (valence, arousal, and dominance).

Each cue corresponds to one column in the dataset. For example, we include one column for each of the EPITOME mechanisms [30] with values {0, 1, 2} denoting {*none*, *weak*, *strong*} displays. Similarly, intent, sentiment, and main subject (*who*) are encoded as different categorical values per column. Furthermore, the utterance length for each interlocutor and binary labels for `mimicry` are included. Finally, each exchange includes the ground-truth `empathy` corresponding to the perceived empathy in 3-level and binary labels.

VII. EMPATHY CLASSIFIER

One of the most difficult tasks for empathy classification is the lack of data [6]. Therefore, we explore the possibility of using a previously successful approach for empathy classification in a multimodal context. For this, we propose using a variation of PBC4cip [18].

A. PBC4emp

We adopt and modify the PBC4cip algorithm implementation in [11]. This is a contrast pattern-based classifier that leverages regularities over data – patterns – that appear significantly more in a class compared to others. These patterns take the form of sets of values for the data features. For example, the pattern {color = red $\wedge$ taste = sweet} will appear more in the class 'fruit' compared to the class 'vegetable' in a hypothetical dataset of agricultural products.

In pattern-based classification, the support of each pattern p into a class c refers to the ratio between the instances covered by the pattern and the total number of instances in the class. Pattern-based classification is carried out by summing the supports for each class of all patterns covering an instance and selecting the class with the highest sum [78]. However, this can lead to bias in imbalanced datasets due to mining algorithms finding more patterns with higher supports for the majority class. To avoid this, PBC4cip [18] calculates weights for the classes based on the imbalance in the training set. During the training phase, class weights are computed according to the following equation:

$$w_c = \frac{(1 - \frac{|c|}{|O|})}{\sum_{p \in P} \text{support}(p, c)} \tag{9}$$

where $|c|$ refers to the total number of instances in class c, $|O|$ refers to the total number of objects in the training set, P the set of all patterns found in the data for class c, and support(p, c) represents the support of the pattern p into class c. The term $(1 - \frac{|c|}{|O|})$ rewards the sum of supports computed for the minority class, while $\sum_{p \in P}$ support(p, c) normalizes the sum of supports in each class regarding all the patterns of the same class [18].

After computing the class weights during the training phase, inference for PBC4cip is the same as the one proposed by other pattern-based classifiers [78]. The supports for patterns that cover an instance o are summed for each of the classes. However, these sums are also multiplied by the class weight w_c. This process is carried out using the following formula:

$$w_{Sum_Supp}(o, c) = w_c * \sum_{\substack{p \in P \\ p \text{ covers } o}} \text{support}(p, c) \tag{10}$$

The term $w_{Sum_Supp}(o, c)$ for each class c is then used to select the output of the classifier. The class with the highest sum is output as the final classification.

The main advantage of PBC4cip is that it performs better than other pattern-based classifiers in imbalanced datasets [18]. Besides, it is also an explainable algorithm, which means that insight can be obtained on the nature of the data and the classification process [6], [29]. This has been previously done to study the nature of empathy by identifying the most common patterns for each class in the training set [11]. We are interested in applying the explainable capabilities of the algorithm to multimodal instances. Therefore, we modify the algorithm to provide explainable information per individual inference.

1) RANKING MODULE

As our main contribution to the PBC4cip algorithm, we propose a novel ranking module that computes the influence of each empathy cue during the classification process. We are able to identify and sort which features contributed more to a given classification. The influence of each empathy cue on the final classification is given by the influence score I_{ec}, which is computed using the following term:

$$I_{ec} = \sum_{\substack{p \in P \\ p \text{ covers } o \\ ec \in p}} support(p, c) \tag{11}$$

where ec represents the empathy cue, p represents a pattern, P is the set of patterns that define the exchange, and

c represents the predicted class. Per cue, we gather the patterns that cover the exchange o and include the cue. Then, we sum up the supports of those patterns for the predicted class [18]. The influence score represents the relevance of a cue per algorithm inference. Since the classification is already computed, we do not consider the class weights for obtaining the relevance scores per each empathy cue. We sort the influence scores to get the top-4 most relevant empathy cues in each inference. Our algorithm outputs the classification label and the top-4 empathy cues with the highest influence scores.

Unlike previous approaches for explainable empathy classification algorithms such as EPITOME [30], our ranking algorithm does not simply highlight portions of a text sequence. Instead, it provides multiple high-level characteristics that contribute to empathy, allowing a more thorough exploration of the multiple dimensions of the concept.

We name our modified implementation the contrast **P**attern-**B**ased **C**lassifier **for emp**athy (PBC4emp). PBC4emp is our base model that predicts empathy per exchange using contextual cues. Then, we augment its capabilities by ensembling it with pre-trained models that increase and complement its original cues with visual data.

B. TEXT-ONLY DATASET

For our training data, we use an extended version of the EmpatheticExchanges (EX) dataset [11]. We adopt their 3-level *Empathy Score* labels. Afterwards, we adapt our 2-level metric. This dataset consisted of 23106 exchanges in dyadic conversations with balanced emotional contexts, empathy labels, and cases of negative [19] and positive empathy [23]. BERT, PBC4emp, and a Random Forest (RF) classifier are used. We selected BERT as a baseline based on the performance of language models for empathy classification [6] and RF as an non-explainable alternative using traditional machine learning, as it has been shown to be useful and even outperforming PBC4cip in this task [11], [29]. All classifiers were trained on 18485 exchanges and tested on 4621. We report results in Table 2.

We observe competitive performance across all three proposed baselines, with results considerably higher than those of the best-performing method reported in the original EX paper for a 3-level empathy metric (PBC with a CEM of 0.597). This performance spike allows us to obtain a new state-of-the-art for empathy classification in text. We make the dataset with our selection for empathy cues available in our repository.

EX is a dataset with diverse conversations between people based on situations that made them feel a certain emotion. As such, we believe that it would allow us to capture the characteristics of empathy regardless of the context. We focus on obtaining an abstract representation of behavioral empathy that can be applied to any context. Therefore, it was necessary to use a diverse series of conversations. Given the reliability of the algorithms on EX, we hypothesized that training a classifier on this data could yield a classifier that managed to predict the empathy between two people even in multimodal and group contexts.

TABLE 2. PBC4emp performance in the EX samples.

2-level					
Model	Acc	CEM	F1	p	r
BERT	0.863	-	0.737	0.778	0.712
PBC4emp	0.834	-	0.743	0.726	0.769
RF	**0.878**	-	**0.774**	**0.805**	**0.753**
3-level					
Model	Acc	CEM	F1	p	r
BERT	0.730	0.758	0.653	0.695	0.629
PBC4emp	0.725	**0.794**	0.732	0.744	0.725
RF	**0.776**	0.788	**0.765**	**0.780**	**0.776**

C. BASELINE MODELS

For empathy classification, previous work has shown that most approaches are BERT-based models trained on separate empathy datasets [6]. Likewise, empathy has been labeled and classified using large language models, obtaining comparable or better results than other deep-learning models [12], [36], [37]. These two approaches have achieved state-of-the-art results. As such, we test PBC4emp against baselines from previous literature: The BERT model fine-tuned for empathy classification using an empathy dataset [11], and GPT-4o with an empathy classification prompt. In addition, we include RF on account of its performance on the EX dataset. Finally, direct comparisons between our approach and text classification algorithms from previous research [5], [28], [30], [32], [33], [35] was not possible due to them using different databases and incompatible metrics for empathy. As such, we use the most promising models tailored for empathy classification using our metrics.

Existing multimodal data for empathy classification [6] were either incompatible or insufficient in size. This highlights the contribution of our research, as we propose a method for incorporating multimodal information without the need to train a new classifier. Nevertheless, we show an implementation of these multimodal classifiers trained on the sparse data of mEX in Section IX.

We fine-tuned BERT using the text from the EX dataset. Meanwhile, we give a prompt to GPT-4o using a similar approach to that in [12], in which we provide a scale, empathy definition, and examples to the model to classify empathy in exchanges. The advantage of GPT-4o was that we could augment its capabilities using additional features. Therefore, we could obtain another baseline for the multimodal cue integration framework proposed.

D. VISUAL CUE INTEGRATION FRAMEWORK

Our video-based cues include three types of mimicry (**1v**, **2v**, and **3v**) as well as arousal and valence (AV) values extracted from participants' facial expressions (**4v**). As shown in Figure 5, we enhance our base model, PBC4emp, by using an integration framework of models that combines aggregation and substitution of empathy cues.

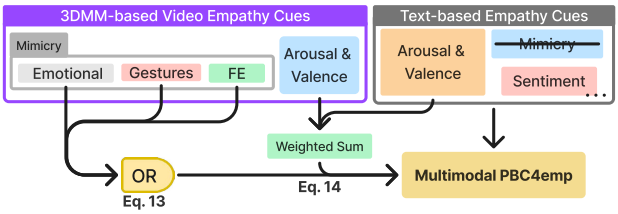


FIGURE 5. Overview of the integration framework that combines text-based and video-based empathy cues (a_{fused} and m_{joint}) for multimodal classification. FE: facial expression.

To capture the presence of video-based mimicry, we replace the original text-based mimicry cue (**5c**) with each of the video-based mimicry cues. In addition, we include a composite video-based signal. This aggregated video-based mimicry is derived from the three binary mimicry classifiers: (**1v**) for facial expression, (**2v**) for head pose, and (**3v**) for emotional mimicry.

Let the individual mimicry outputs be defined as:

$$m_{\mathbf{1v}},\ m_{\mathbf{2v}},\ m_{\mathbf{3v}} \in \{0, 1\} \tag{12}$$

We then define the aggregated video-based mimicry cue as a logical OR operation over the three types:

$$m_{\mathbf{joint}} = m_{\mathbf{1v}} \vee m_{\mathbf{2v}} \vee m_{\mathbf{3v}} \tag{13}$$

where $m_{\mathbf{joint}} = 1$ if at least one of the mimicry types is present, and $m_{\mathbf{joint}} = 0$ only if all $m_{\mathbf{1v}}, m_{\mathbf{2v}}, m_{\mathbf{3v}}$ are zero.

We explore effects of the composite cue $m_{\mathbf{joint}}$, $m_{\mathbf{1v}}$, $m_{\mathbf{2v}}$, and $m_{\mathbf{3v}}$ by replacing the original text-based mimicry input (**5c**) in our model. Each mimicry type is selected individually and used as a video-derived social signal in subsequent classification tasks.

Additionally, we enhance the original text-based arousal and valence values (**3c**) by incorporating information from the user's facial expressions captured via (**4v**). Let $\mathbf{a_{3c}}, \mathbf{a_{4v}} \in [-1, 1]^2$ denote the AV vectors from text and video modalities, respectively. We compute a fused AV representation through a weighted sum:

$$\mathbf{a_{fused}} = (1 - \lambda) \cdot \mathbf{a_{3c}} + \lambda \cdot \mathbf{a_{4v}}, \quad \lambda \in [0, 1] \tag{14}$$

where λ is a tunable hyperparameter that controls the contribution of each modality. The resulting vector $\mathbf{a_{fused}}$ is then used as the final AV input to our base model.

For GPT-4o, we include an explanation in the prompt regarding the presence or absence of video-based mimicry. Additionally, we directly incorporate the video-based cues into the prompt. Readers can review our prompt in the Appendix.

VIII. EVALUATION

The first step in our evaluation was to select the appropriate parameters for our integration framework method for AV vectors and video-based mimicry. To achieve this, we conducted empirical tests. We evaluated the results of our classifier using the tunable λ parameter from Equation 14, adjusting it in iterative increments of 0.1 between 0 and 1. The parameter

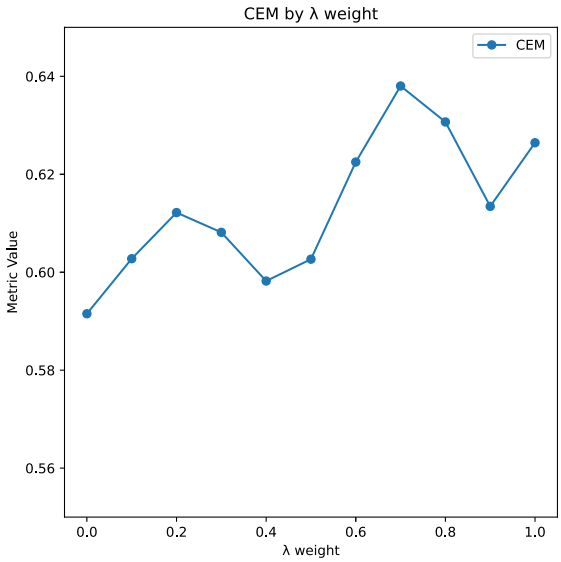


FIGURE 6. Iterative process for λ hyperparameter tuning based on the CEM score.

selection process is illustrated in Figure 6. We then refined the search using smaller increments of 0.01 between 0.60 and 0.8, obtaining the best performance at $\lambda = 0.70$. Therefore, we selected this value for all subsequent evaluations.

For video-based mimicry, we explored four approaches: using only **1v**, only **2v**, only **3v**, and the aggregated mimicry feature (m_{joint}). However, we did not observe any significant differences among the individual implementations of mimicry in PBC4emp, and its overall effect appeared to negatively impact classifier performance. Nevertheless, we remained interested in examining its interaction with AV multimodal vectors.

Given the tunable hyperparameter λ, the proportion was set to 30:70 for the text-based and video cues, respectively. We observed a notable change in the distribution of AV values after integration. Figure 7 presents violin plots showing the distributions of arousal and valence values derived from text and video sources, as well as after performing their weighted combination.

We report the performance of both the 2-level and 3-level versions of our model, alongside the baseline models, evaluated on mEX. For conciseness, we include only the results obtained with the best-performing λ value. We selected facial mimicry (**2v**) for integration, as no significant differences were observed among the various mimicry implementations and it seemed to obtain the best results. All results are summarized in Table 3. The PBC4emp model achieved the highest performance across most metrics in both 2-level and 3-level evaluations. Moreover, both RF and PBC4emp demonstrated improved performance when video-based empathy cues were incorporated.

We also conducted McNemar's pairwise test to assess the statistical significance of performance differences between models [79]. Table 4 presents all model comparisons that

TABLE 3. Models performance on the mEX dataset. For each column, the best result is bolded, and the second best is underlined. *t* = text-only, *av* = arousal and valence, and *m* = mimicry.

		2-level					3-level				
Model		Acc	CEM	F1	p	r	Acc	CEM	F1	p	r
BERT	*t*	0.5581	-	0.4508	0.6613	0.5444	0.2968	0.5235	0.2198	0.3667	0.3446
RF	*t*	0.5613	-	0.4406	**0.7392**	0.5469	0.3129	0.5242	0.2147	0.4542	0.3709
	t+m	0.5613	-	0.4406	**0.7392**	0.5469	0.3129	0.5248	0.2150	0.4486	0.3709
	t+av	0.5613	-	0.4489	0.6957	0.5473	0.3258	0.5308	0.2383	**0.5216**	0.3838
	t+av+m	0.5581	-	0.4428	0.6908	0.5440	0.3226	0.5297	0.2346	0.4876	0.3815
GPT-4o	*t*	0.6710	-	**0.6582**	0.6890	0.6654	0.4516	**0.6544**	0.4327	0.4900	0.4406
	t+m	0.6032	-	0.5693	0.6315	0.5948	0.3903	0.6199	0.3860	0.4538	0.4082
	t+av	0.6516	-	0.6371	0.6685	0.6458	0.4355	0.6474	0.3934	0.5069	0.4287
	t+av+m	0.6258	-	0.6019	0.6499	0.6185	0.4000	0.6304	0.3405	0.4665	0.4049
PBC4emp	*t*	0.6129	-	0.5679	0.6650	0.6031	0.4032	0.5993	0.3667	0.4397	0.4305
	t+m	0.6161	-	0.5763	0.6611	0.6069	0.4097	0.6035	0.3761	0.4392	0.4331
	t+av	**0.6774**	-	0.6568	0.7152	**0.6702**	**0.4710**	0.6468	**0.4400**	0.4833	**0.4758**
	t+av+m	0.5161	-	0.4240	0.5091	0.5037	0.2839	0.5009	0.2327	0.3158	0.3287

TABLE 4. McNemar pairwise comparisons (p-values) for 2-level and 3-level settings. Only pairs with $p < 0.05$ and involving a variant of PBC4emp are shown. *t* = text-only, *av* = arousal and valence, *m* = mimicry.

Model A	Model B	2-level	3-level
PBC4emp *t*	RF *t*	0.02374	0.00024
	RF *m*	0.02374	0.00024
	RF *t+av*	0.02064	0.00212
	RF *t+av+m*	0.01469	0.00178
	PBC4emp *t+av*	0.01417	0.01455
	PBC4emp *t+av+m*	0.00119	0.00011
	BERT	0.04719	0.00027
PBC4emp *m*	RF *t*	0.02506	0.00023
	RF *m*	0.02506	0.00023
	RF *t+av*	0.02227	0.00178
	RF *t+av+m*	0.01621	0.00149
	PBC4emp *t+av*	0.01911	0.02334
	PBC4emp *t+av+m*	0.00063	0.00003
	BERT	0.04216	0.00019
PBC4emp *t+av*	RF *t*	0.00005	0.00000
	RF *m*	0.00006	0.00000
	RF *t+av*	0.00002	0.00000
	RF *t+av+m*	0.00001	0.00000
	PBC4emp *t+av+m*	0.00000	0.00000
	BERT	0.00008	0.00000
	GPT-4o *m*	0.02859	–
PBC4emp *t+av+m*	GPT-4o *av*	0.00005	0.00047
	GPT-4o *m*	0.00585	0.00875
	GPT-4o *av+m*	0.00066	0.00909
	GPT-4o *t*	0.00000	0.00007

involve PBC4emp and that exhibited significant differences the $p < 0.05$ threshold.

A. FREQUENCY OF EMPATHY CUES

We present the frequency in which all classification features appear in the top-4 most influential cues through mEX. We decided to use the top-4 as the length of the utterances was overwhelmingly present in the top-3 influential features, with utterance length present in 270 out of 310 exchanges. This is consistent with findings that lengthy responses are preferred in empathetic utterances [80]. Since this was already observed, we exclude length features in our analysis. We present the results in Figure 8.

TABLE 5. Most influential features for the example exchange classified by PBC4emp. Number denotes the ranking: 1) is best, given by the highest I_{ec}. *Class* refers to the classification obtained using the model.

Model	Listener Cues	Speaker Cues	Class
t	**1)** length = 10 **2)** emotional_reaction = 0 **4)** dominance ≈ 0.17	**3)** length = 18	2/2
t+av	**1)** length = 10 **4)** arousal ≈ 0.00	**2)** dominance ≈ −0.05 **3)** valence ≈ 0.05	1/2

B. EXAMPLE OF CLASSIFICATION AND EXPLAINABILITY

To illustrate the behavior of our proposed model, we present an example of the output of an empathy classification instance in Table 5. In it, we show the most relevant features extracted by our explainability algorithm. The exchange was initially misclassified as empathy level 2/2 using only text-based features (*t*), but was correctly classified after incorporating AV cues from video (*t+av*). This example demonstrates how multimodal integration enhances the model's interpretation of empathic context while providing insight into the cues driving its decision. The exchange is as follows:

> **Speaker:** [talking about an encounter with a spider] yeah, they're like very nasty and if they, like, bite you, I heard some can actually kill you.
> **Listener:** I will immediately run away if I saw a spider.
>
> **Empathy class:** 1 / 2

Interpretation: As shown in Table 5, the text-only model assigns greater importance to features such as utterance length, the fact that there is no emotional reaction presented by the listener and that they have control over the conversation (dominance). In contrast, the multimodal (*t+av*) model captures additional affective information from both participants. Greater importance is assigned to the speaker's

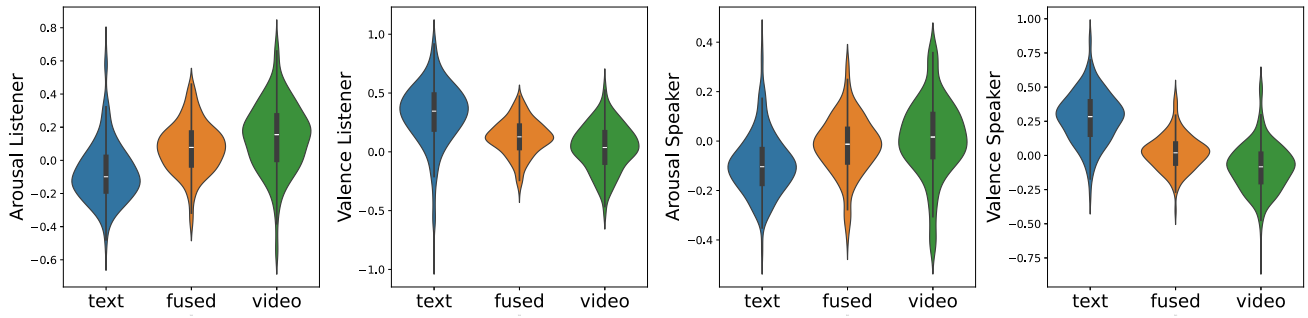


FIGURE 7. Violin plots showing the original arousal and valence values extracted from text and from video using our 3DMM-based method, as well as the combined results obtained through our ensemble approach.

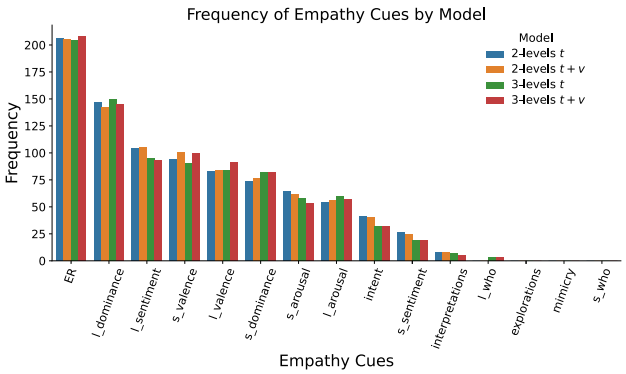


FIGURE 8. Frequency of empathy cues among the top-4 influential features for our proposed model in 3-level and 2-level classifications, using text alone or combined with arousal, valence, and mimicry cues from video. s_ = speaker, l_ = listener, ER = emotional reaction.

sense of control and negative affect. Combined with the short utterance length and neutral engagement (arousal), this makes the prediction of a low empathy score intuitive. This discrepancy between verbal and nonverbal affective cues contributes to a more accurate classification of low empathy.

IX. MULTIMODAL BASELINES

Although the primary intention of the mEX dataset is to serve as a test set for evaluating pre-trained models, in this section we additionally report results from multiple multimodal baselines trained and evaluated directly on mEX. The dataset is split into training and test sets using an 0.8–0.2 ratio (we refer readers to our repository for the training and test datasets).

We selected baseline model families inspired by the best-performing approaches reported in recent empathy datasets, namely OMG-Empathy [4] and EmpathicStories++ [12]. These datasets are among the most representative and recent benchmarks for computational empathy.

OMG-Empathy focuses on predicting the affective impact of a story on a listener using continuous annotations. Its strongest baseline uses a Support Vector Machine (SVM) regressor [81], highlighting the importance of temporal modeling and regression-based learning for empathy prediction.

EmpathicStories++ addresses the task of predicting how empathic a person feels toward stories told by a robot about other people. The best-performing models reported include LLMs (GPT-4), a transformer-based language model based on BART [82], and sequence-based multimodal models such as LSTMs, with performance varying depending on the interaction phase and evaluation metric.

Based on these findings, we include the following multimodal baselines in our evaluation: (1) an LSTM-based model to capture temporal dynamics, (2) a transformer-based language model to represent high-level semantic content (BERT), (3) an SVM classifier since it outperformed RF in OMG-Empathy [83], and (4) an LLM (GPT-4o). In addition, we include (5) a random forest (RF) method, as it was the best-performing approach in our tests on the EX dataset.

The baselines include both top-down and bottom-up approaches. The top-down methods are RF, SVM, and PBC4emp. For these models, we first extract salient empathy-related cues from different modalities and then use these cues as input for classification. To ensure a fair comparison, all top-down methods rely on the same set of empathy cues proposed for PBC4emp (see Section V). In contrast, the bottom-up methods are LSTM and BERT. Rather than using handcrafted cue representations, these models receive the utterance directly, augmented with multimodal information. Specifically, AV and mimicry signals are incorporated into the input as additional features encoded as special tokens or one-hot representations.

A. EVALUATION

Table 6 reports the results of all models trained and evaluated on **mEX** for both binary and 3-level empathy classification. Overall, **PBC4emp** achieves the strongest performance across most metrics in both settings. In particular, it consistently performs best when combining mimicry cues with arousal and valence information extracted from video.

X. ONLINE SETTING

As a proof of concept for using empathy classifiers in online settings, we tested PBC4emp in our multigroup application, The Talking Room [16]. This setup involves two groups of

TABLE 6. Multimodal models trained on the MEX dataset. For each column, the best result is **bolded**, and the second best is underlined. *t* = text-only, *av* = arousal and valence, and *m* = mimicry.

		2-level					3-level				
Model		Acc	CEM	F1	p	r	Acc	CEM	F1	p	r
LSTM	$t + m$	0.4769	-	0.4475	0.4749	0.4806	0.2615	0.4576	0.2289	0.2759	0.2528
	$t + av$	0.5846	-	0.5782	0.5875	0.5829	0.4615	0.5967	0.3755	0.4023	0.3778
	$t + av + m$	0.5077	-	0.4050	0.5042	0.5014	0.4769	0.5328	0.2452	0.2250	0.3215
BERT	$t + m$	0.5077	-	0.5048	0.5093	0.5090	0.4154	0.5728	0.3719	0.3692	0.3927
	$t + av$	0.5692	-	0.5642	0.5750	0.5710	0.4154	0.6035	0.3962	0.4133	0.4055
	$t + av + m$	0.5385	-	0.5192	0.5503	0.5417	0.3077	0.5680	0.2670	0.3247	0.3201
RF	$t + m$	0.6615	-	0.6576	0.6667	0.6600	0.5538	0.6600	0.4609	0.4642	0.4668
	$t + av$	0.6769	-	0.6720	0.6850	0.6752	0.5692	0.6705	0.4719	0.4782	0.4769
	$t + av + m$	0.6462	-	0.6407	0.6525	0.6444	0.5538	0.6571	0.4628	0.4699	0.4668
SVM	$t + m$	0.6462	-	0.6289	0.6728	0.6430	**0.6000**	0.6702	0.5093	**0.5680**	0.5024
	$t + av$	0.6154	-	0.6061	0.6237	0.6132	0.5692	0.6585	0.4967	0.5238	0.4906
	$t + av + m$	0.6308	-	0.6199	0.6432	0.6283	0.5846	0.6664	0.5065	0.5408	0.5007
GPT-4o	$t + m$	0.5077	-	0.4872	0.5132	0.5109	0.3692	0.5998	0.3777	0.4495	0.4026
	$t + av$	0.4769	-	0.4667	0.4772	0.4792	0.3077	0.5572	0.2988	0.4621	0.3095
	$t + av + m$	0.5077	-	0.4932	0.5119	0.5104	0.3077	0.5705	0.2858	0.5635	0.3148
PBC4emp (ours)	$t + m$	0.6154	-	0.6095	0.6260	0.6174	0.4615	0.6381	0.4365	0.4476	0.4420
	$t + av$	0.6923	-	0.6917	0.6954	0.6932	0.5538	0.6853	0.4932	0.4914	0.5058
	$t + av + m$	**0.7077**	-	**0.7066**	**0.7128**	**0.7088**	0.5846	**0.7045**	**0.5331**	0.5379	**0.5481**



FIGURE 9. The Talking Room setting, showing two groups of children interacting with each other through an online video call.

three or four high school-age children in separate locations. Each group interacts with a Haru robot and a screen that connects them to the other group for ≈ 40 minute sessions. The two Haru robots coordinate to mediate the group conversation, as shown in Figure 9. The study received ethical approval from both our institution and the participating schools. We do not have permission to share the children's interaction data. For more details on this application, please refer to our website.[2]

The participants included three children from Japan, four from Australia, three from Namibia, and three from Denmark in the Talking Room sessions. All children voluntarily agreed to participate in the study and provided informed consent, either independently or with parental approval. We collected 83 exchanges between the children over two sessions. Session 1 (S1) included Namibia and Denmark (43 exchanges), while Session 2 (S2) involved Japan and Australia (40 exchanges). This was the children's first session interacting with peers from the other country. The children were between 10 and 15 years old.

[2]https://mypersonalrobots.org/pilot

TABLE 7. Model performance in the talking room scenario. For each column, the best result is **bolded** and the second best is underlined.

	3-level				
Model	Acc	CEM	F1	p	r
BERT	0.397	0.584	0.300	0.416	0.451
RF	0.421	0.591	0.293	0.406	0.457
GPT-4o	**0.710**	**0.708**	0.402	0.438	**0.416**
PBC4emp	0.662	0.680	**0.424**	**0.458**	0.411

The 3-ES label distribution of the exchanges is 53 for level 1, 26 for level 2, and 4 for level 3. We display the results of the baselines and our empathy model in Table 7. The session-wise results using our 3-ES model are as follows: means of $\mu_{\mathbf{S1}} = 1.3$ and $\mu_{\mathbf{S2}} = 1.41$, with variances $\sigma^2_{\mathbf{S1}} = 0.42$ and $\sigma^2_{\mathbf{S2}} = 0.39$.

XI. LIMITATIONS

Our work has several limitations that should be considered when interpreting the results. First, a key limitation concerns the composition of the participants used for data collection, which includes four speakers and 17 listeners. In particular, the mEX dataset includes four speakers, three of whom are male. This may introduce bias in how emotional situations are narrated and expressed, potentially influencing the distribution and manifestation of behavioral empathy cues. However, to mitigate this limitation, all speakers followed predefined emotional contexts adapted from the ED dataset, which spans a wide range of affective situations and emotional categories. By grounding the conversations in diverse, standardized emotional contexts, we aimed to reduce speaker-specific effects and encourage variability in

emotional content across exchanges. Nevertheless, consistent narrative tendencies may still affect listener responses, which could limit generalization to more diverse populations and real-world settings.

Second, the multicultural composition of the participants constitutes an additional limitation. Empathy is known to be culturally dependent, with norms governing emotional expression, nonverbal behavior, and empathic responses varying across cultures. Consequently, interactions between individuals from diverse cultural backgrounds may introduce additional variability that is difficult to capture with a single, unified model. Likewise, language might have an effect on the expression of empathy, with participants being able to speak but not all of them having english as their native language. This cultural heterogeneity may lead to underfitting in certain contexts or mask culture-specific empathy cues that would otherwise be salient in more homogeneous settings. It is important to note that the effects and findings of our research are limited to behavioral empathy as conceptualized in the english language and constrained to those cultural constructs.

Third, behavioral empathy is an inherently ambiguous and subjective construct, which makes reliable annotation challenging. This difficulty is reflected in the relatively low inter-annotator agreement observed in our dataset, with Cohen's κ values of 0.23 and 0.38 for the three-level and two-level annotations, respectively. While many prior empathy datasets do not report inter-annotator agreement—often because they rely on self-assessment rather than external observation—annotation variability remains a known challenge in empathy research. These results suggest that annotation instability is a characteristic of behavioral empathy datasets rather than a limitation unique to our data.

Fourth, another limitation is the size of our dataset, mEX. Although it is sufficient to train deep learning models, it is not large enough to support the same level of generalization as models trained on substantially larger datasets. For example, when we evaluate on EX a PBC4emp model trained on **mEX** using 4,616 samples (577 per emotion), it achieves a three-level accuracy of 0.2743, CEM of 0.5288, precision of 0.3824, recall of 0.3748, and F1 of 0.2688. These results remain close to chance level, and they are also worse than those of the EX-trained PBC4emp model evaluated on **mEX**. This gap is consistent with the expected effect of dataset scale on model robustness and transferability. However, the main purpose of mEX is not to serve as a large-scale training corpus, but to provide a benchmark for evaluating the generalization of pretrained models originally developed on EmpatheticExchanges, the largest publicly available dataset for behavioral empathy. From this perspective, our findings are encouraging, as they show that these models retain competitive performance when tested on mEX. At the same time, the limited size of mEX makes it difficult to determine whether it fully captures the diversity of scenarios encountered in real-world empathic interactions. Future work should therefore examine whether the findings reported in this paper remain consistent on larger and more diverse datasets. Nevertheless, mEX represents an important step forward: to the best of our knowledge, it is one of the first datasets that enables the evaluation of whether behavioral empathy classifiers can transfer to dyadic interactions involving both visual and audio contact, moving the field closer to more realistic and ecologically valid settings.

Future work involving participants from more homogeneous cultural backgrounds, multiple speakers with different characteristics or anonymous identities, and a larger dataset will be necessary to validate whether our multimodal feature integration framework consistently preserves performance across contexts.

XII. DISCUSSION AND FUTURE WORK

The results of this research demonstrate the feasibility of augmenting text-based empathy classification with multimodal features. We integrated multimodal cues into a pretrained, pattern-based model and applied it to behavioral empathy classification in both face-to-face and online group-call settings. Overall, the results show that PBC4emp transfers well across datasets and remains competitive under changes in conversational medium and context, including differences in vocabulary, common expressions, interjections, and the online group setting used in The Talking Room. In addition, our framework provides an interpretable way to examine the contribution of individual cues within a top-down classification model.

We found that nonverbal cues affect performance, but their impact depends strongly on the cue type and on the training setting. On mEX, visual arousal–valence (AV) cues consistently improved PBC4emp over the text-only variant when the model was trained on EX, yielding the best 2-level and 3-level results for this model. In contrast, mimicry alone provided only limited gains, and combining AV with mimicry reduced performance for the EX-trained PBC4emp model. However, when the model was trained directly on mEX, the best performance was obtained by combining text, AV, and mimicry cues. This contrast suggests that the usefulness of mimicry depends on the compatibility between the training distribution and the target domain, rather than being uniformly beneficial or harmful. More broadly, AV cues had a stronger and more consistent effect than mimicry across most baselines, while our cue-frequency analysis showed that mimicry was rarely selected among the most influential features.

PBC4emp augmented with multimodal cues also remained highly competitive with the other baselines. For EX-trained models evaluated on mEX, PBC4emp with text and AV cues achieved the strongest overall results among the tested variants, especially in F1 and recall, although GPT-4o remained competitive in some metrics such as CEM. For models trained directly on mEX, PBC4emp achieved the best overall 2-level performance when text, AV, and mimicry were combined, and it also obtained the strongest 3-level CEM, F1, and recall. By contrast, GPT-4o did not show consistent

gains from multimodal augmentation, and in several cases its text-only version remained stronger than its multimodal variants. This finding suggests that simply adding more information to a large language model is not sufficient. Possible reasons include interference between cues, excessive prompt complexity, or a mismatch between the prompt design and the structure needed to integrate multimodal information. These results highlight the importance of principled fusion strategies rather than assuming that additional cues will always improve performance.

When we analyzed the most influential features, we found that emotional reaction (ER) was the most informative cue beyond utterance length across both 2-level and 3-level settings. Listener dominance and valence-related cues also appeared frequently, whereas arousal cues appeared less often and mimicry was almost never selected among the top influential features. This pattern suggests that empathy classification in our framework relies more on cues related to affective alignment and the listener's expressive stance than on simple behavioral mirroring. It also helps explain why AV cues were more useful than mimicry in most of our experiments.

Empathy is known to increase with familiarity and perceived similarity between individuals [7], which may help explain the generally low empathy levels observed in The Talking Room, where participants interacted for the first time. In this setting, PBC4emp achieved the highest F1 and precision, while GPT-4o achieved the highest accuracy, CEM, and recall. These results suggest that PBC4emp remains robust in more natural and class-imbalanced scenarios, where low-empathy responses are likely to be common. One possible reason is that the pattern-based design of PBC4emp is well suited to skewed behavioral data, which is a realistic property of real-world empathy classification tasks.

Accurate empathy classifiers could provide affective computing researchers with practical tools for the automatic analysis of empathic behavior. As discussed in prior work [6], this objective has already attracted attention, but our results suggest a more specific opportunity: pretrained and interpretable classifiers such as PBC4emp can be transferred across related domains while still providing meaningful performance and cue-level explanations. This creates useful possibilities for applications in which labeled data are limited, such as cross-cultural analysis of low-empathy interactions, training for mental health professionals, or educational settings in which systems highlight empathic and non-empathic responses. More broadly, the consistent transfer from EX to mEX and the competitive results in The Talking Room indicate that pretrained behavioral empathy models may support practical downstream use without requiring full retraining for every new interaction setting.

Finally, our annotation process showed that empathy classification is inherently difficult even for expert annotators. We observed a substantial discrepancy between the first annotation round and the final labels, and this discrepancy was larger for the 3-level scheme than for the binary one.

TABLE 8. McNemar pairwise comparisons (p-values) of all classification algorithms besides PBC4emp for 2-level and 3-level settings. Only pairs with $p < 0.05$ are shown. *t* = text-only, *av* = arousal and valence, *m* = mimicry.

Model A	Model B	2-level	3-level
RF *t*	GPT-4o *av*	0.00536	0.00727
	GPT-4o *av*+*m*	0.03365	–
	GPT-4o *t*	0.00044	0.00108
RF *m*	GPT-4o *av*	0.00488	0.00697
	GPT-4o *av*+*m*	0.03589	–
	GPT-4o *t*	0.00044	0.00108
RF *t*+*av*	GPT-4o *av*	0.00360	0.01553
	GPT-4o *av*+*m*	0.02930	–
	GPT-4o *t*	0.00037	0.00328
RF *t*+*av*+*m*	GPT-4o *av*	0.00268	0.01291
	GPT-4o *av*+*m*	0.02092	–
	GPT-4o *t*	0.00027	0.00262
BERT	GPT-4o *av*	0.00580	0.00237
	GPT-4o *m*	–	0.02830
	GPT-4o *av*+*m*	0.04229	0.02681
	GPT-4o *t*	0.00056	0.00026
GPT-4o *av*	GPT-4o *m*	0.04550	–
GPT-4o *m*	GPT-4o *t*	0.00700	–

This suggests that empathy becomes harder to label reliably as the number of target categories increases. During the discussion before the second annotation round, the main disagreement concerned whether the listener expressed a resonant emotion or an unrelated one. Once the definition was refined with concrete examples of hypothetical empathic behaviors, both annotators reached full consensus. These observations highlight the subjectivity of empathy perception and the need for clearer operational definitions and stronger annotation protocols in future work. One useful direction would be to familiarize annotators with well-defined binary cases first, especially cases in which the listener expresses either a resonant emotion or an unrelated emotional response.

XIII. CONCLUSION

We proposed *PBC4emp*, a method for extending a pretrained text-based empathy classifier with multimodal cues without retraining the original model. Our results show that PBC4emp transfers effectively across EX, mEX, and *The Talking Room*, remaining competitive across different interaction settings. Visual arousal–valence cues consistently improved performance in cross-dataset evaluation, whereas mimicry showed a less consistent effect. In addition, PBC4emp provides an interpretable structure that helps reveal which empathy cues contribute most to classification.

We also introduced *mEX*, a multimodal dataset of dyadic face-to-face conversations, and evaluated our models in *The Talking Room*, an online group interaction scenario. Together, these results suggest that pretrained behavioral empathy classifiers can retain useful performance across different conversational settings, while highlighting the value of principled multimodal fusion and interpretable cue analysis for future affective computing applications.

APPENDIX. PROMPTS FOR GPT-4o

We include the prompts for GPT-4o used for benchmarking. The prompt presented is for classification using transcriptions, VA vectors, and pose mimicry.

Rate the behavioral empathy displayed by the listener per exchange. Our definition of empathy supports three dimensions in a hierarchical order: Affective, Cognitive, and Behavioral. Affective empathy means having emotional congruence with another person (same or similar emotion); Cognitive empathy is to understand the feelings and emotions of another and why they feel that way; Behavioral empathy is to communicate in a manner that demonstrates cognitive and affective empathy. Note that high empathy includes expressions of interest for the other person's emotional state or their situation.

Also consider the additional metrics we extracted from video: Arousal (range: -1 to 1): Reflects the level of emotional intensity of the speaker and listener, with -1 being very calm and 1 being highly aroused. Valence (range: -1 to 1): Indicates the emotional valence (positive or negative) of the speaker and listener, with -1 being very negative and 1 being very positive. Pose mimicry (binary: 1 or 0): Indicates whether the listener mimics the speaker's head movement (1 for yes, 0 for no).

Give your answer on this scale: 1 - little to no empathy, 2 - somewhat empathetic, and 3-empathetic.

An example of a pair of exchanges with empathy level 1: speaker: There was strange noise from the kitchen at night. listener: Did you have your alarm on speaker: Yes, but it didnt ring. I was spooked but it turned out I had left the window open and it was the wind. listener: Oh wow, I would have thought maybe an animal had gotten in.

An example of a pair of exchanges with empathy level 2: speaker: I was babysitting a few days ago and my cousin kept throwing her food on the floor. listener: That must have been frustrating. What did you do? speaker: I kept cleaning it up and put down a bunch of towels around her to protect the floor. I think it comes with the territory, she gets what she wants. listener: That made me laugh. And you are a great babysitter.

An example of a pair of exchanges with empathy level 3: speaker: My dad knew that I have recently hit a rough patch with my finances and gave me 2 grand. I cannot thank him enough. listener: How kind of him! I bet that is going to help you out a lot! speaker: It really is going to help a lot. I feel so blessed! listener: Dads are the best, what would we do without them?

Answer format: Rate: . . . Reason: . . .

Other prompts used for this purpose had the following differences:

- Modified rating scale to be *Give your answer on this scale: 1 - no empathy, 2 - empathetic. .*
- Modified example with level 3 to have level 2 during binary classification.
- Eliminated the condition to consider additional metrics.

APPENDIX. McNemar PAIRWISE RESULTS FOR OTHER MODELS

We include the pairwise results from McNemar tests in the mEX dataset for all other classification algorithms. We were focused in the performance of our proposed classifier. However, we disclose all statistically significant comparisons between all baselines in the multimodal dataset, as it is important for understanding the relative performance between algorithm families and among different implementations.

We observe that all variations of GPT-4o, besides the one using only mimicry as a multimodal factor, outperform significantly all implementations of RF and BERT in binary classification. Likewise, there is a statistically significant difference between text-only GPT-4o and all other models, with the exception being the augmented implementations of the same LLM with arousal and valence values.

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